

Gear and Bearing Design for Rotor–Motor Geared Drive

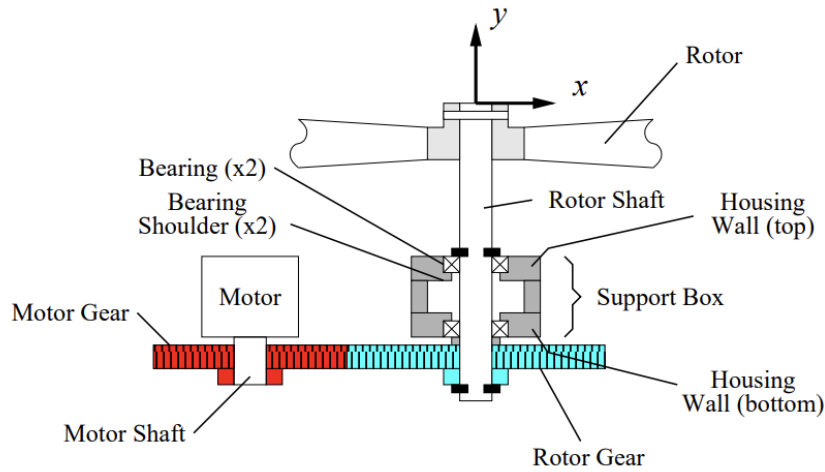
Key design outputs

Symbol	Description	Value
d_{shaft}	Shaft diameter, motor & rotor	25 mm (0.984 in)
N_m	Motor gear teeth	30
N_r	Rotor gear teeth	60
$i = N_r/N_m$	Final reduction ratio (motor:rotor)	2.00:1
P_d	Diametral pitch	10 teeth/in
m	Normal module	2.54 mm
C	Nominal center distance	4.50 in
d_b	Bearing bore	25 mm
D_b	Bearing outer diameter	52 mm
B	Bearing width	15 mm
$C_{0,\text{req}}$	Required static load rating (incl. FoS)	≈ 200 lbf
C_0	Selected static rating (6205-2RS)	1550 lbf
$C_{10,\text{req}}$	Required basic dynamic rating (incl. FoS, thrust included in P)	$\approx 2.5 \times 10^3$ lbf
C_{10}	Selected basic dynamic rating (6205-2RS)	2650 lbf
t_{wall}	Axial wall thickness at bearing seat (bearing width + shoulder)	0.80 in ($\approx 20\text{mm}$)
t_{rad}	Radial wall thickness around bearing bore	0.50 in ($\approx 12.7\text{mm}$)
t_{shoulder}	Housing shoulder height behind bearing	0.18 in ($\approx 4.6\text{mm}$)
D_{housing}	Housing bore for bearing outer ring (Timken one-piece housing fit)	52.000–52.019 mm (loose)
d_{journal}	Shaft journal for inner ring (Timken “rotating IR, normal loads”)	25.011–25.017 mm (tight)
$d_{\text{sh,ID}}$	Shoulder through-hole ID and tolerance	25.5 mm nominal, e.g. ± 0.1 mm
F_t	Tangential gear-tooth load at rotor gear	157 lbf
F_r	Radial gear-tooth load at rotor gear	57.3 lbf
F_a	Axial gear-tooth load (spur gear)	0 lbf
F_{min}	Minimum face width to meet required FoS (contact)	0.92 in
F	Catalog face width (selected)	1.00 in
BHN_{min}	Minimum Brinell hardness for gears	≥ 250 BHN
–	Recommended material grade	Q&T alloy steel (e.g. AISI 4140/4340)
Q_v	Recommended AGMA quality factor	$Q_v \approx 11$

Bill of Materials (drive and support box)

Qty	Item	Vendor/PN	Description	Unit Cost (\$)
2	Bearing	McMaster 6153K79	6205-2RS SS deep-groove ball, 25 mm bore	61.99
1	Gear (rotor)	McMaster 6325K9	10 DP, 60T, mach. bore, 1144 steel	125.02
1	Gear (motor)	McMaster 6325K39	10 DP, 30T, mach. bore, 1144 steel	111.29
Estimated total cost				360.29

Layout and Assumptions



Drawing not to scale
(support beam not shown)

Figure 1: Geared rotor-motor layout

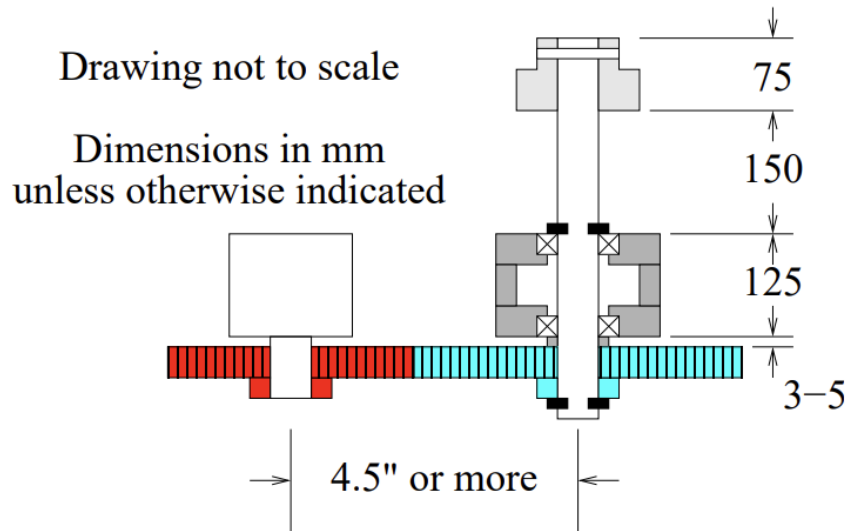


Figure 2: Geared rotor-motor dimensions (not to scale)

The belt drive is replaced by a spur gear mesh between the motor shaft and rotor shaft. The rotor shaft is supported by two identical ball bearings in the top and bottom walls of a 6061-T6 aluminum support box. A spacer between the rotor gear and the bottom bearing prevents rubbing between the gear hub and the box. Features above the top wall (propeller hub, etc.) are unchanged and do not affect the bearing and gear sizing. Later justified, the chosen parts are:

- 2× Stainless Steel Ball Bearing, sealed, trade no. 6205-2RS, 25 mm shaft (*McMaster-Carr* 6153K79, \$61.99 each).
- 1× spur gear, 10 pitch, 60 teeth, 14.5° PA, machinable bore (*McMaster-Carr* 6325K9, \$125.02).

- 1× spur gear, 10 pitch, 30 teeth, 14.5° PA, machinable bore (*McMaster-Carr* 6325K39, \$111.29).

Key assumptions and given data:

1. Both shafts are steel with diameters ≥ 0.75 in in all regions. To match the selected gears and metric bearings, the design uses a shaft diameter in the gear and bearing seats,

$$d_{\text{motor}} = d_{\text{rotor}} = \boxed{25 \text{ mm} \approx 0.984 \text{ in}},$$

which exceeds the minimum 0.75 in requirement. Both McMaster gears, even though in imperial units, have machinable hubs and are able to be bored to 25 mm = 0.984 in.¹

2. Motor power and speed range:

$$P = 16.75 \text{ kW} = 22.5 \text{ hp}, \quad 4500 \leq n_m \leq 6000 \text{ rpm}.$$

Rotor operating requirements:

$$n_r = 3000 \text{ rpm}, \quad T_r = 53.317 \text{ N m} = 471.89 \text{ in lbf}.$$

Rotor thrust and weight:

$$F_{\text{thrust}} = 170.6 \text{ lbf (upward)}, \quad W_R = 6.75 \text{ lbf (downward)}.$$

Rotor weight is small compared to gear forces and thrust and is neglected in bearing radial sizing.

3. Shaft center spacing must satisfy

$$C \geq 4.5 \text{ in}$$

to clear the motor body and support box.

4. The rotor shaft is carried by identical deep-groove ball bearings (type 6205-2RS, stainless, sealed) in the box walls. Catalog geometry and ratings for the selected McMaster-Carr bearing (6153K79) are:

$$d_b = 25 \text{ mm}, \quad D_b = 52 \text{ mm}, \quad B = 15 \text{ mm},$$

$$C_{10} = 2650 \text{ lbf}, \quad C_0 = 1550 \text{ lbf}.$$

The distance between bearing centerlines (box height) is

$$L = 125 \text{ mm} = 4.921 \text{ in}.$$

The rotor gear is below the bottom wall; a spacer of thickness t_{sp} prevents contact with the box.

5. Design life requirement: at least 1200 h of operation at $n_r = 3000$ rpm. Bearings are designed on an L_{10} basis with factor of safety $n = 1.2$ on *load*.²

¹McMater-Car Gears Part No. 6325K9 and 6325K39

²Bearing L_{10} life for ball bearings: $L_{10} = (C_{10}/P)^3 10^6$ rev (ball bearings have life exponent $a = 3$); see Shigley, Eq. (11-10).

6. Bearings must be corrosion-resistant and sealed against water (not just shielded).
7. Gears must be spur gears sourced from McMaster–Carr. The selected gears are 10-pitch, 30/60 teeth, 1144 carbon steel spur gears with catalog 14.5° pressure angle; for analysis they are treated as 20° full-depth involute gears so that standard AGMA geometry and strength factors can be applied.³ The supplier provides allowable bending and contact stresses

$$S_t = 35 \text{ ksi}, \quad S_c = 110 \text{ ksi},$$

already adjusted for the required reliability.⁴

8. No undercutting is permitted. For 20° full-depth spur gears, this requires $N_{\text{pinion}} \gtrsim 18$ teeth,⁵ which is satisfied by the selected 30-tooth pinion.
9. AGMA bending and contact stress models (Shigley, Ch. 14) are used with target gear factor of safety

$$n_{\text{gear}} \geq 1.5$$

for both bending and contact.⁶

10. The housing walls are 6061–T6 aluminum with yield strength $S_y \approx 35$ ksi. The walls and shoulders are sized so that local stresses in the aluminum remain comfortably below yield.

Bearing Reactions

A free-body diagram (FBD) of the rotor shaft includes:

- Tangential gear-tooth force F_t at the rotor gear pitch circle, acting at $x = 0$.
- Radial gear-tooth force F_r (toward the gear center) at the same axial plane.
- Bottom bearing reactions $R_{A,t}$ and $R_{A,r}$ at $x = L_1$.
- Top bearing reactions $R_{B,t}$ and $R_{B,r}$ at $x = L_2 = L_1 + L$.
- Axial thrust F_{thrust} transmitted through the retaining ring, gear, spacer, and bottom bearing.

With the gear below the lower bearing, the gear forces act *outside* the between-bearings distance. Taking the gear midplane as $x = 0$, the bottom bearing at $x = L_1$, and the top bearing at $x = L_2 = L_1 + L$, static equilibrium for either transverse plane (tangential or radial) gives

$$\begin{aligned} R_A + R_B &= F, \\ \sum M_A &= 0 \Rightarrow R_B L - F L_1 = 0 \Rightarrow R_B = F \frac{L_1}{L}, \\ R_A &= F - R_B = F \left(1 - \frac{L_1}{L} \right), \end{aligned}$$

where F is F_t in the tangential plane and F_r in the radial plane. These expressions are used below to compute bearing loads.

³Use of 20° full-depth geometry for strength factors follows standard AGMA practice and the design specification; see Shigley, Sec. 13–3 and 14–5.

⁴Allowable bending and contact stresses including a reliability factor K_R are discussed in Shigley, Sec. 14–8.

⁵Minimum teeth to avoid undercut for 20° full-depth involute spur gears: Shigley, Sec. 13–3.

⁶AGMA bending and contact stress forms: Shigley, Sec. 14–6.

Gearbox Kinematics and Layout

Gear ratio, tooth counts, and components

To keep the rotor at or below 3000 rpm over the motor range and reach 3000 rpm at the top end $n_m = 6000$ rpm, the chosen speed ratio is

$$i = \frac{n_m}{n_r} = \frac{6000}{3000} = 2.0.$$

Define the reduction ratio in terms of teeth as

$$i = \frac{N_{\text{rotor}}}{N_{\text{motor}}}.$$

The selected McMaster gears are:

- **Motor gear (pinion):** 30 teeth, 10-pitch, machinable bore, 1144 carbon steel. McMaster–Carr 6325K39.
- **Rotor gear:** 60 teeth, 10-pitch, machinable bore, 1144 carbon steel. McMaster–Carr 6325K9. The hub is sized for shafts in the 7/8–1.5 in range and will be bored to 25 mm.

Thus,

$$N_m = 30, \quad N_r = 60, \quad i = \frac{N_r}{N_m} = \boxed{2.0}.$$

Over the full motor range, this yields rotor speeds

$$n_r = \frac{n_m}{i} = \begin{cases} 2250 \text{ rpm} & (n_m = 4500 \text{ rpm}), \\ 3000 \text{ rpm} & (n_m = 6000 \text{ rpm}), \end{cases}$$

so the desired rotor speed of 3000 rpm is achieved at the top end of the motor range, and the rotor never exceeds its nominal limit.

Diametral pitch and center distance

Using diametral pitch P_d (teeth per inch), the pitch diameters and center distance are⁷

$$d_m = \frac{N_m}{P_d}, \quad d_r = \frac{N_r}{P_d}, \quad C = \frac{d_m + d_r}{2} = \frac{N_m + N_r}{2P_d}.$$

The chosen McMaster gears are 10-pitch:

$$P_d = \boxed{10 \text{ teeth/in}},$$

so

$$d_m = \frac{30}{10} = 3.00 \text{ in}, \quad d_r = \frac{60}{10} = 6.00 \text{ in},$$

and the resulting center distance is

$$C = \frac{3.00 + 6.00}{2} = \boxed{4.50 \text{ in}}.$$

This exactly meets the minimum center-distance requirement $C \geq 4.5$ in, and both gears are standard 10-pitch spur gears with the desired tooth counts.

⁷Gear pitch geometry using diametral pitch: Shigley, Sec. 13-2.

Gear Tooth Forces

The rotor torque T_r is transmitted through the rotor gear of pitch diameter $d_r = 6.00$ in. The tangential force at the pitch circle is

$$F_t = \frac{2T_r}{d_r} = \frac{2(471.89)}{6.00} \approx \boxed{157 \text{ lbf}}.$$

With a 20° design pressure angle for analysis,

$$F_r = F_t \tan \phi, \quad \phi = 20^\circ,$$

so

$$F_r \approx 157 \tan 20^\circ \approx \boxed{57.3 \text{ lbf}}.$$

There is no axial component for spur gears, so $F_a = 0$ at the gear mesh.

The same tangential and radial forces act on the motor gear (equal and opposite), and both gears see the same pitch-line velocity. At $n_r = 3000$ rpm,

$$v = \frac{\pi d_r n_r}{12} = \frac{\pi(6.00)(3000)}{12} \approx \boxed{4.7 \times 10^3 \text{ ft/min}},$$

which is used to determine the dynamic factor K_v . (The rotor torque T_r was given and is consistent with the 22.5 hp motor at 6000 rpm and a 2:1 gear ratio, so there was no need to find T_r through power.)

Bearing Loads from Gear Forces and Thrust

Axial geometry along the rotor shaft

Take the gear tooth force to act in the midplane of the rotor gear. From that plane up to the bottom bearing centerline we approximate

$$L_1 = \frac{F}{2} + t_{\text{sp}} + \frac{t_{\text{wall}}}{2},$$

where:

- Catalog gear face width $F = \boxed{1.00 \text{ in}}$.
- Spacer thickness $t_{\text{sp}} = \boxed{4 \text{ mm} = 0.157 \text{ in}}$ (within the required 3–5 mm range).
- Total axial bottom-wall thickness (bearing width + shoulder) chosen as

$$t_{\text{wall}} \approx B + t_{\text{shoulder}} \approx 0.59 + 0.18 \approx \boxed{0.80 \text{ in}},$$

consistent with the 6205 bearing width.

Thus

$$L_1 \approx 0.50 + 0.157 + 0.40 \approx \boxed{1.06 \text{ in}}.$$

The distance between the two bearing centerlines is $L = 4.921$ in, so the top bearing is at axial coordinate

$$L_2 = L_1 + L \approx 1.06 + 4.92 = 5.98 \text{ in}$$

above the gear midplane.

Radial reactions at the bearings

Using the equilibrium results for a transverse load F acting at $x = 0$:

$$R_B = F \frac{L_1}{L}, \quad R_A = F \left(1 - \frac{L_1}{L}\right).$$

Tangential plane. With $F = F_t = 157$ lbf,

$$R_{B,t} = 157 \frac{1.06}{4.921} \approx 33.9 \text{ lbf}, \quad R_{A,t} \approx 157 - 33.9 = 123 \text{ lbf}.$$

Radial plane. With $F = F_r = 57.3$ lbf,

$$R_{B,r} = 57.3 \frac{1.06}{4.921} \approx 12.3 \text{ lbf}, \quad R_{A,r} \approx 57.3 - 12.3 = 45.0 \text{ lbf}.$$

Resultant radial loads. The resultant radial loads at each bearing are

$$F_A = \sqrt{R_{A,t}^2 + R_{A,r}^2} \approx \sqrt{123^2 + 45.0^2} \approx \boxed{131 \text{ lbf}},$$

$$F_B = \sqrt{R_{B,t}^2 + R_{B,r}^2} \approx \sqrt{33.9^2 + 12.3^2} \approx \boxed{36 \text{ lbf}}.$$

The bottom bearing (A) clearly governs the radial design.

Axial load in the rotor bearings

The rotor thrust $F_{\text{thrust}} = 170.6$ lbf is carried by retaining rings on the shaft that react against the support box via the gear and spacer stack. For a conservative bearing design, the entire thrust is assumed to pass through the bottom bearing inner and outer races:

$$F_{a,A} \approx \boxed{170.6 \text{ lbf}}, \quad F_{a,B} \approx 0.$$

Bearing Design: Static and Dynamic Ratings

The bottom bearing carries combined radial and axial load and sets the required bearing ratings. The inner ring rotates with the shaft, so $V = 1$ in the dynamic load equations.

Equivalent dynamic load

For deep-groove ball bearings under combined radial and axial load, Shigley follows the ISO/ABMA practice of defining an equivalent dynamic load

$$P = XF_r + YF_a$$

with coefficients X and Y depending on F_a/F_r and F_a/C_0 .⁸ For deep-groove ball bearings with $F_a/F_r \lesssim 1$ the ISO tables give $X \approx 0.56$ and $Y \approx 1.63$.⁹

⁸Equivalent dynamic load for ball bearings: Shigley, *Mechanical Engineering Design*, Sec. 11–3 and Table 11–1.

⁹Coefficients X and Y for deep-groove ball bearings are tabulated in ISO 281 and in most manufacturer catalogs; Shigley reproduces typical values in Table 11–1.

For the bottom bearing,

$$F_r = F_A \approx 131 \text{ lbf}, \quad F_a = F_{a,A} \approx 170.6 \text{ lbf},$$

so

$$\frac{F_a}{F_r} \approx 1.30, \quad \frac{F_a}{C_0} \approx \frac{170.6}{1550} \approx 0.11.$$

Using the deep-groove ball-bearing coefficients $X = 0.56$, $Y = 1.63$,

$$P = XF_r + YF_a \approx 0.56(131) + 1.63(170.6) \approx 73 + 278 \approx \boxed{P \approx 351 \text{ lbf}}.$$

This P is larger than radial-only F_A and therefore gives a conservative basis for the life calculation.

C_{10} **and** L_{10}

With L_{10} in revolutions,¹⁰

$$L_{10} = \left(\frac{C_{10}}{P} \right)^3 10^6.$$

For the required life

$$L_{10,\text{req}} = (1200 \text{ h})(60 \text{ min/h})(3000 \text{ rev/min}) = 2.16 \times 10^8 \text{ rev},$$

so

$$\left(\frac{L_{10,\text{req}}}{10^6} \right)^{1/3} = 6.$$

The basic dynamic rating needed *including thrust* is therefore

$$C_{10,\text{req}} = P \cdot 6 \approx 6(351) \approx 2106 \text{ lbf}.$$

Applying the design factor $n = 1.2$ on load gives

$$C_{10,\text{design}} = nC_{10,\text{req}} \approx 1.2(2106) \approx \boxed{2527 \text{ lbf}}.$$

The selected 6205-2RS bearing has

$$C_{10,\text{actual}} = 2650 \text{ lbf},$$

so the margin on C_{10} is small but positive:

$$\frac{C_{10,\text{actual}}}{C_{10,\text{design}}} \approx \frac{2650}{2527} \approx \boxed{1.05}.$$

In other words, the chosen bearing just meets the required dynamic rating once thrust is fully included and the factor of safety on load has been applied.

For reference, the actual life at the combined load P is

$$L_{10,\text{actual}} = \left(\frac{2650}{351} \right)^3 10^6 \approx 3.7 \times 10^8 \text{ rev} \approx 2.05 \times 10^3 \text{ h at 3000 rpm},$$

which is still more than an order of magnitude above the 1200 h requirement.

¹⁰Bearing life for ball bearings: Shigley, Eq. (11-10), with exponent $a = 3$.

Static equivalent load and required C_0 (with thrust)

For ball bearings under combined loading the static equivalent load is¹¹

$$P_0 = 0.6F_r + 0.5F_a.$$

With the radial load $F_r = 131$ lbf and the axial load $F_a = 170.6$ lbf,

$$P_0 = 0.6(131) + 0.5(170.6) \approx 78.6 + 85.3 \approx 164 \text{ lbf.}$$

Including the static factor of safety $n_s = 1.2$,

$$C_{0,\text{design}} = n_s P_0 \approx 1.2(164) \approx 197 \text{ lbf.}$$

The selected bearing has $C_0 = 1550$ lbf, so

$$n_{0,\text{actual}} = \frac{C_0}{C_{0,\text{design}}} \approx \frac{1550}{197} \approx 7.9.$$

Static capacity is therefore not limiting.

Wall and Shoulder Thickness

For layout and strength calculations, using the geometry of the chosen 6205-2RS bearing:

$$d_b = 25 \text{ mm} \approx 0.984 \text{ in}, \quad D_b = 52 \text{ mm} \approx 2.05 \text{ in}, \quad B = 15 \text{ mm} \approx 0.59 \text{ in}.$$

The housing is 6061-T6 aluminum with yield strength $S_y \approx 35$ ksi.

Wall thickness: radial support check

The bottom bearing transmits the radial reaction F_A into the housing through the outer race. A simple and conservative way to size the wall is to require that the compressive stress in the bearing seat region of the wall remain well below yield.

If radial thickness of the housing around the bore is t_{rad} (distance from bearing OD to the outside of the wall, we can approximate the local contact patch between the bearing outer race and the wall as having area

$$A_{\text{wall}} \approx B t_{\text{rad}},$$

so the nominal compressive stress in this region is

$$\sigma_{c,\text{wall}} = \frac{F_A}{A_{\text{wall}}} = \frac{F_A}{B t_{\text{rad}}}.$$

Choosing a preliminary radial thickness

$$t_{\text{rad}} = \boxed{0.50 \text{ in}},$$

gives

$$A_{\text{wall}} = (0.59)(0.50) \approx 0.30 \text{ in}^2,$$

¹¹Static equivalent load: Shigley, Eq. (11-11).

$$\sigma_{c,\text{wall}} = \frac{131}{0.30} \approx 4.4 \times 10^2 \text{ psi} = 0.44 \text{ ksi.}$$

For 6061-T6 with $S_y \approx 35$ ksi, even with a conservative design factor $n_{\text{wall}} = 2$ we would allow

$$\sigma_{\text{allow,wall}} = \frac{S_y}{n_{\text{wall}}} = \frac{35}{2} \approx 17.5 \text{ ksi.}$$

The actual factor of safety in compression is therefore

$$n_{c,\text{wall}} = \frac{S_y}{\sigma_{c,\text{wall}}} \approx \frac{35}{0.44} \approx 80.$$

Thus, a 0.50 in radial wall thickness keeps the local compressive stress in the bearing seat region orders of magnitude below yield. Additional contact stresses from a light press fit are also well below yield.

Shoulder thickness and axial wall thickness

The axial thrust $F_{a,A}$ is transmitted from the bearing outer race into the housing via a shoulder behind the outer race. We can model the shoulder as a rectangular ring section of thickness t_{shoulder} (axial height) and width B . The average compressive stress in the shoulder cross-section is

$$\sigma_{c,\text{sh}} = \frac{F_{a,A}}{A_{\text{sh}}}, \quad A_{\text{sh}} = B t_{\text{shoulder}}.$$

Using the guideline that the shoulder height should be at least about $0.3B$ for deep-groove ball bearings,¹² we choose

$$t_{\text{shoulder}} \approx 0.3B \approx 0.3(0.59) \approx \boxed{0.18 \text{ in}}.$$

Then

$$A_{\text{sh}} = (0.59)(0.18) \approx 0.11 \text{ in}^2,$$

$$\sigma_{c,\text{sh}} = \frac{170.6}{0.11} \approx 1.6 \text{ ksi.}$$

We can also check the average shear stress along the circumference at the root of the shoulder:

$$\tau_{\text{sh}} \approx \frac{F_{a,A}}{\pi D_b t_{\text{shoulder}}} = \frac{170.6}{\pi(2.05)(0.18)} \approx 150 \text{ psi.}$$

Both compressive and shear stresses are tiny compared to $S_y = 35$ ksi; with $n \approx 2$ on yield, the shoulder has a large margin.

Per standard housing practice, the total axial wall thickness is the sum of the bearing width and shoulder thickness¹³ With $B \approx 0.59$ in and $t_{\text{shoulder}} \approx 0.18$ in,

$$t_{\text{wall}} = B + t_{\text{shoulder}} \approx 0.59 + 0.18 \approx \boxed{0.77 \text{ in}},$$

which we round to 0.80 in for layout. Being greater than the preliminary 0.18 in, shoulder area is greater, which only decreases contact stress (so no need to redo calculations).

¹²Typical shoulder-height recommendations: Shigley, Sec. 11-2, housing and mounting discussion.

¹³See, e.g., SKF *Rolling Bearings* catalog, housing design section, which recommends that the abutment shoulder and adjacent wall be at least as wide as the bearing width for rigid support.

Fits and Tolerances

To avoid creep of the inner ring and ensure adequate support of the outer ring without overstressing the 6061-T6 housing, shaft and housing fits are chosen directly from Timken’s “Bearing Fitting Practice” tool for part number 6205-2RS (ABEC 1 radial deep-groove ball bearing, 25x52x15 mm)¹⁴. The tables used are:

- *Inner ring tolerances and fits*, ABEC 1, 25 mm bore: inner ring bore nominal 25.000 mm with tolerance -0.010 to 0.000 mm; for a rotating inner ring under normal radial load, Timken recommends shaft diameters in the range 25.011–25.002 mm (“0.021 tight” to “0.002 tight” fit).
- *Outer ring tolerances and fits*, ABEC 1, 52 mm OD: outer ring OD nominal 52.000 mm with tolerance -0.013 to 0.000 mm; for a stationary outer ring in a one-piece aluminum housing under all loads, Timken recommends housing bores in the range 52.019–52.000 mm (“0.032 loose” to “0.000 loose” fit).

Based on these manufacturer tables, the design adopts:

- **Shaft (inner-ring fit, heavy loads/shock):**

Inner ring bore: 24.990–25.000 mm, Shaft OD: 25.017–25.008 mm,

corresponding to a resultant interference of 0.027–0.008 mm (Timken “heavy load, shock” recommendation). This is a light press fit that prevents inner-ring creep under the combined radial and axial load while remaining machinable in a steel shaft.

- **Housing bore (outer-ring fit, one-piece housing):**

Outer ring OD: 51.987–52.000 mm, Housing bore: 52.019–52.000 mm,

giving a resultant clearance of 0.032–0.000 mm (Timken “all loads, one-piece housing” recommendation). This slight clearance/transition fit is specifically intended to account for press-fit effects, thermal expansion differences, and housing material while still preventing outer-ring creep under normal loads.

- **Shoulder through-hole:** The shoulder through-hole is given a nominal diameter

$$d_{\text{sh,ID}} \approx 25.5 \text{ mm } (\approx 1.00 \text{ in}),$$

comfortably larger than the 25 mm shaft to provide assembly clearance while the shoulder face supports the bearing outer ring across the full width B .

These fits are not arbitrary; they are taken directly from Timken’s fitting recommendations for the selected 6205-2RS bearing and therefore already include the manufacturer’s assumptions on press-fit, material, and loading.

Gear Strength: Bending and Contact

The motor gear (30 teeth) is the pinion and, due to smaller size, is more critical for both bending and contact fatigue than the larger rotor gear. Strength checks are therefore based on the pinion.

¹⁴<https://engineering.timken.com/engineering-tool/bearing-fitting-practice>, Part No. 6205-2RS

Lewis form factor and size factor K_s

For a 30-tooth, 20° full-depth involute spur gear, Shigley’s Lewis form-factor table gives¹⁵

$$Y \approx \boxed{0.35}.$$

Using the size-factor correlation¹⁶

$$K_s = 1.192 \left(\frac{F\sqrt{Y}}{P_d} \right)^{0.0535},$$

with $F = 1.00$ in and $P_d = 10$, we get

$$K_s \approx \boxed{1.03}.$$

Overload and dynamic factors K_o , K_v

An overload factor

$$K_o = \boxed{1.25}$$

is used to represent an electric motor driving an aerodynamic rotor (approximately uniform power source with moderate transient loading).¹⁷

The AGMA dynamic factor K_v is obtained from the Barth equation¹⁸

$$B = 0.25(12 - Q_v)^{2/3}, \quad A = 50 + 56(1 - B), \quad K_v = \left(\frac{A + \sqrt{V}}{A} \right)^B,$$

where Q_v is the AGMA quality number and V is pitch-line velocity (ft/min). For the McMaster parts chosen (10-pitch, 30T and 60T 1144 carbon steel spur gears), the original manufacturer and AGMA quality grade are not published; Shigley’s discussion of the Barth velocity factor K_v notes that typical AGMA quality numbers for commercial 20° spur gears range from about $Q_v \approx 6$ (roughly cut, commercial quality) to $Q_v \approx 12$ (precision ground).¹⁹ For this design:

- power and speed are moderate,
- smooth rotor operation is desired (but not aerospace-level precision), and
- the gears are open-mounted with standard machining tolerances.

It is therefore reasonable to estimate a mid-to-high AGMA quality number, and the analysis adopts

$$\boxed{Q_v \approx 11}$$

as a representative quality level for a well-made, accurately cut spur gear set. This choice is within the “precision” range illustrated by Shigley and leads to the dynamic factor

$$K_v \approx 1.15$$

¹⁵Lewis form factors for 20° full-depth spur gears: Shigley, Table 14–2.

¹⁶Size factor K_s for gears: Shigley, Eq. (14–26).

¹⁷Typical overload factors for motor-driven machinery: Shigley, Table 14–6.

¹⁸Velocity factor K_v : Shigley, Eqs. (14–27)–(14–29).

¹⁹See Shigley, Fig. 14–9 and the accompanying text for typical Q_v ranges and their influence on K_v .

used in the bending and contact-stress calculations, and gives

$$B = 0.25(1)^{2/3} = 0.25, \quad A = 50 + 56(1 - 0.25) = 92.$$

At $V \approx 4.7 \times 10^3$ ft/min,

$$K_v = \left(\frac{92 + \sqrt{4700}}{92} \right)^{0.25} \approx \boxed{1.15}.$$

Load-distribution factor K_m

Using the AGMA load-distribution factor for uncrowned, open gearing²⁰

$$K_m = 1 + C_{mc}(C_{pf}C_{pm} + C_{ma}C_e),$$

we have:

- Uncrowned teeth $\Rightarrow C_{mc} = 1$.

- Face-width factor C_{pf} for narrow face width:

$$\frac{F}{10d_p} = \frac{1.0}{10(3.0)} \approx 0.033 < 0.05 \Rightarrow \frac{F}{10d_p} = 0.05,$$

$$\text{and for } \frac{F}{10d_p} \leq 1, C_{pf} = \frac{F}{10d_p} - 0.025 = 0.05 - 0.025 = \boxed{0.025}.$$

- Because the gear is not straddle-mounted, it is good practice to conservatively adopt $C_{pm} = \boxed{1.1}$ to account for greater misalignment than the straddle case.
- For open gearing, the surface-misalignment factor is

$$C_{ma} = A + BF + CF^2,$$

with $A = 0.247$, $B = 0.0167$, and $C = -0.000756$. For $F = 1.00$ in,

$$C_{ma} = 0.247 + 0.0167(1.00) - 0.000756(1.00)^2 \approx \boxed{0.263}.$$

- No deliberate lapping or special crowning $\Rightarrow C_e = 1$.

Thus

$$K_m = 1 + (0.025)(1.1) + (0.263)(1) \approx \boxed{1.29}.$$

Geometry factors J and I

The AGMA bending geometry factor J is taken from the spur-gear charts (Fig. 14-6) for a 30-tooth pinion meshing with a 60-tooth gear, both 20° full-depth. Reading the chart gives

$$J_p \approx \boxed{0.33}.$$

For contact stress, the pitting-resistance geometry factor I for external spur gears is²¹

$$I = \frac{\cos \phi \sin \phi}{2} \frac{m_G}{m_G + 1},$$

with $\phi = 20^\circ$ and gear ratio $m_G = N_g/N_p = 60/30 = 2$. Thus

$$I = \frac{\cos 20^\circ \sin 20^\circ}{2} \frac{2}{2 + 1} \approx \boxed{0.107}.$$

²⁰Load-distribution factor K_m : Shigley, Eqs. (14-30)–(14-35).

²¹Contact geometry factor I for external spur gears: Shigley, Eq. (14-23).

Elastic coefficient C_p and surface condition factor C_f

For steel-on-steel contacts with $E \approx 30 \times 10^6$ psi and $\nu \approx 0.30$, Table 14–8 gives the elastic coefficient²²

$$C_p = \boxed{2300 \sqrt{\text{psi}}}.$$

We assume a ground finish on both gears, so the surface condition factor is taken as

$$C_f = 1.0.$$

Bending strength and minimum face width

The AGMA bending stress equation (U.S. units) for the pinion is²³

$$\sigma_b = \frac{W_t K_o K_v K_s K_m K_B P_d}{F J},$$

with $K_B \approx 1$ for a solid gear rim.

For this design:

- $W_t = 157$ lbf,
- $P_d = 10$ teeth/in,
- $K_o = 1.25$,
- $K_v \approx 1.15$,
- $K_s \approx 1.03$,
- $K_m \approx 1.29$,
- $J \approx 0.33$,
- $K_B \approx 1$,
- $F = 1.00$ in (catalog face width).

Thus

$$\sigma_b = \frac{157(1.25)(1.15)(1.03)(1.29)(10)}{1.00(0.33)} \approx \boxed{9.1 \text{ ksi}}.$$

The allowable bending stress for steel gears is $S_t = 35$ ksi; the bending safety factor is therefore

$$n_{\text{bend}} = \frac{S_t}{\sigma_b} \approx \frac{35}{9.1} \approx \boxed{3.8},$$

well above the required $n_{\text{gear}} = 1.5$.

Solving for the minimum face width required by bending gives

$$F_{\text{min},b} = \frac{W_t K_o K_v K_s K_m K_B P_d}{J S_t / n_{\text{gear}}} \approx \boxed{0.39 \text{ in}}.$$

Thus, bending strength places only a mild lower bound on face width; any realistic catalog face width ($\gtrsim 0.5$ in) easily satisfies the bending requirement. In our case, both gears have a face width of 1 in, easily exceeding the minimum bound.

²²Elastic coefficient C_p : Shigley, Table 14–8.

²³AGMA bending stress form for spur gears in U.S. units: Shigley, Sec. 14–6.

Contact stress and minimum face width

The AGMA contact-stress form (U.S. customary) is²⁴

$$\sigma_c = C_p \sqrt{\frac{W_t K_o K_v K_s K_m C_f}{F d_p I}}.$$

For the pinion:

- $W_t = 157 \text{ lbf}$,
- $d_p = d_m = 3.00 \text{ in}$,
- $C_p = 2300 \sqrt{\text{psi}}$,
- $K_o = 1.25$, $K_v \approx 1.15$, $K_s \approx 1.03$, $K_m \approx 1.29$,
- $I \approx 0.107$, $C_f = 1.0$.

For $F = 1.00 \text{ in}$,

$$\sigma_c = 2300 \sqrt{\frac{157(1.25)(1.15)(1.03)(1.29)(1.0)}{(1.00)(3.00)(0.107)}} \approx \boxed{70 \text{ ksi}}.$$

The allowable contact stress is $S_c = 110 \text{ ksi}$, so the contact safety factor is

$$n_{\text{contact}} = \frac{S_c}{\sigma_c} \approx \frac{110}{70} \approx \boxed{1.6},$$

slightly above the required 1.5.

Solving $\sigma_c \leq S_c/n_{\text{gear}}$ for F gives the minimum face width required by contact fatigue:

$$F_{\text{min},c} = \frac{W_t K_o K_v K_s K_m C_f C_p^2}{d_p I (S_c/n_{\text{gear}})^2} \approx \boxed{0.92 \text{ in}}.$$

Thus, contact stress *governs* the face-width requirement. The catalog face width $F = 1.00 \text{ in}$ is slightly above this minimum, giving $n_{\text{contact}} \approx 1.6$ and satisfying the design target.

Material Hardness and AGMA Quality Recommendation

Minimum Brinell hardness and material grade

The supplier's allowables $S_t = 35 \text{ ksi}$ and $S_c = 110 \text{ ksi}$ (already including reliability) are consistent with medium-carbon alloy steels in the mid-hardness range²⁵. Empirical correlations between hardness and strength for steels indicate that through-hardened steels with Brinell hardness in the range

$$\text{BHN} \approx 250\text{--}300$$

have sufficient root-bending and contact fatigue strength to support these allowables after AGMA corrections.²⁶

A suitable material specification for a new supplier would therefore be:

²⁴ AGMA contact stress form: Shigley, Eq. (14-21).

²⁵ Hardness–strength correlations and gear-strength: Shigley, Table 2-2.

²⁶ Hardness–strength correlations and gear-strength discussion: Shigley, Sec. 14-7 and Table 2-2.

- Material: quenched-and-tempered alloy steel (e.g. AISI 4140 or 4340).
- Minimum surface hardness: $\boxed{\text{BHN} \geq 250}$, with a target range of 250–300.

Because the given AGMA allowables are fatigue-based for long life, a design that meets the bending and contact criteria above also remains comfortably below the static yield strength of such steels.

AGMA quality factor (quality number Q_v)

The gears are specified as high-precision, ground-tooth components. In AGMA terms this corresponds to quality levels $Q_v \approx 10$ –12 (precision ground)²⁷, which in turn supports a relatively low dynamic factor K_v at $v \approx 4700$ ft/min.²⁸

Given:

- power and speed are moderate,
- smooth rotor operation is desired (but not aerospace-level precision), and
- the gears are open-mounted with standard machining tolerances.

a reasonable recommendation to any gear supplier is

$$\boxed{Q_v \approx 11},$$

which was used in the K_v calculation above and is typical for high-quality ground spur gears.

²⁷Load-distribution factor K_m : Shigley, Eqs. (14–30)–(14–35).

²⁸AGMA quality levels and their effect on K_v : Shigley, discussion following Eq. (14–27).